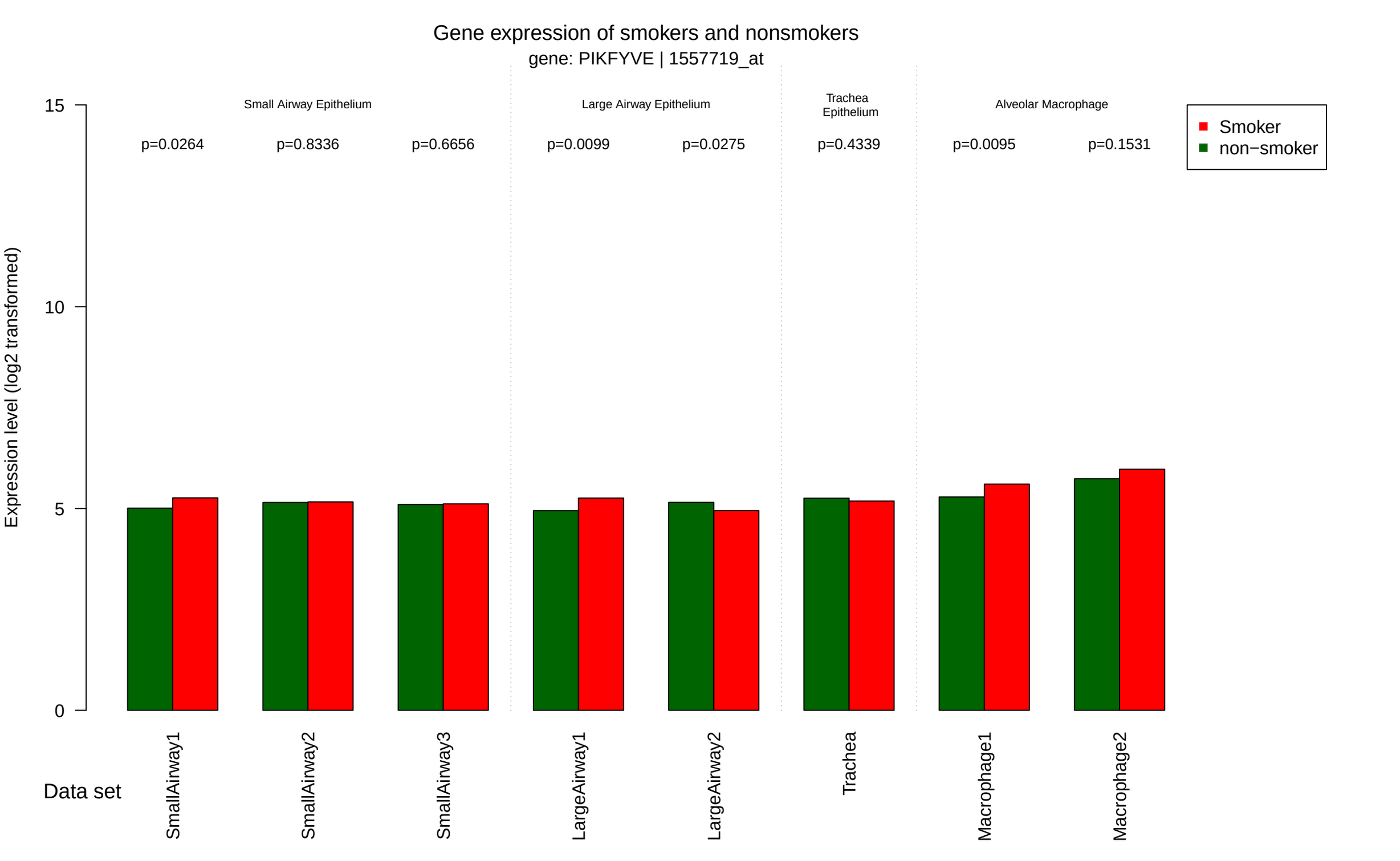
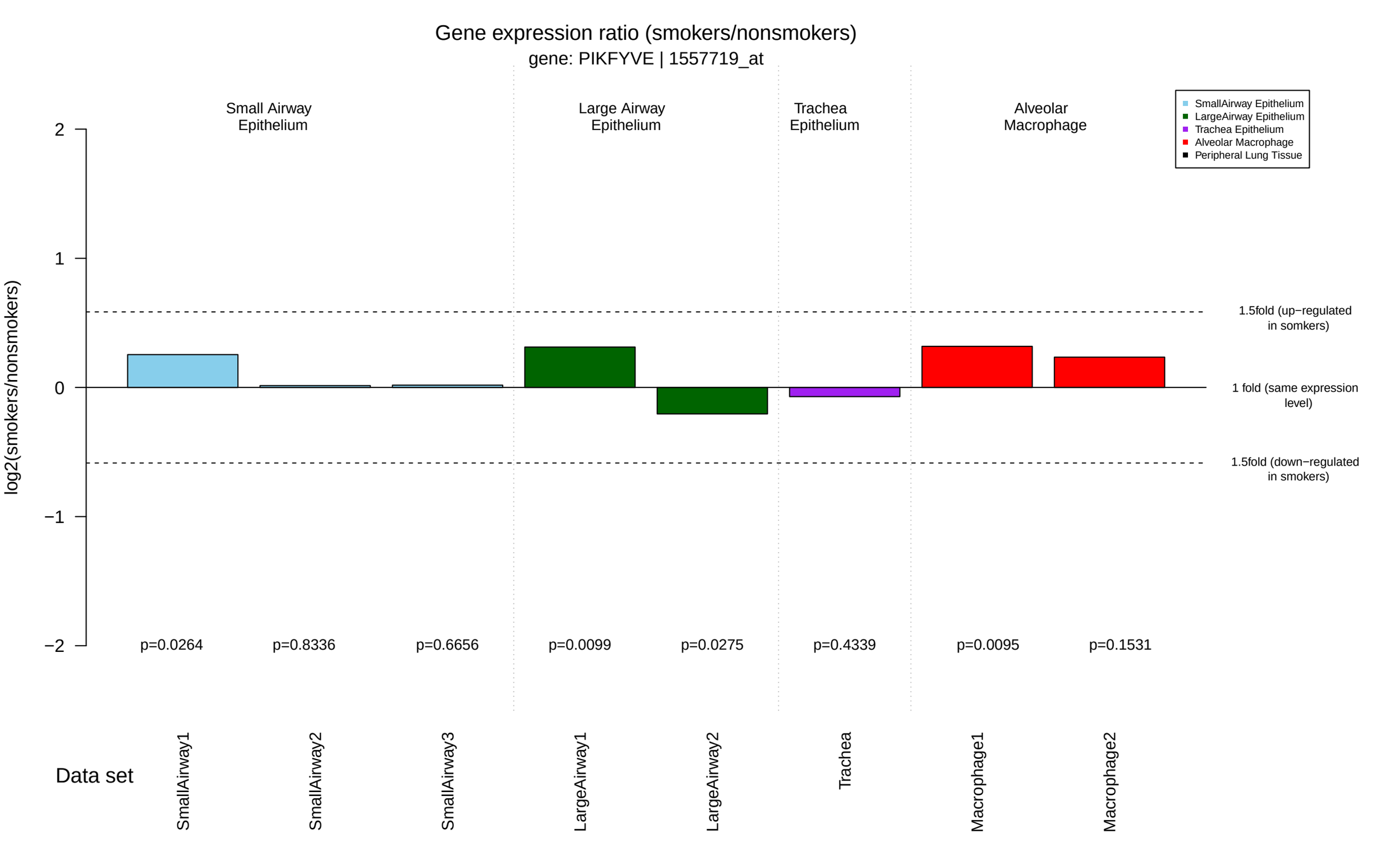
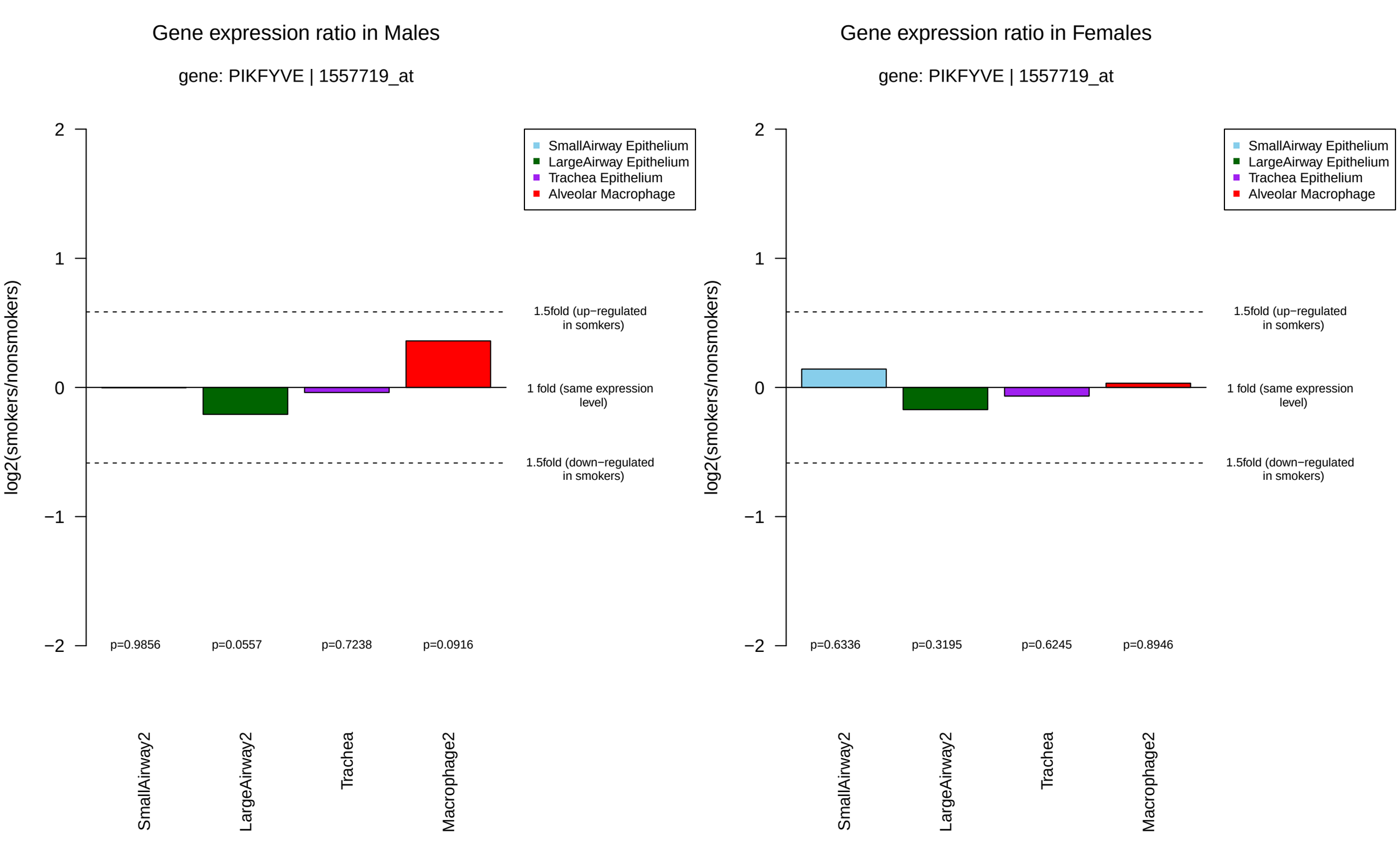
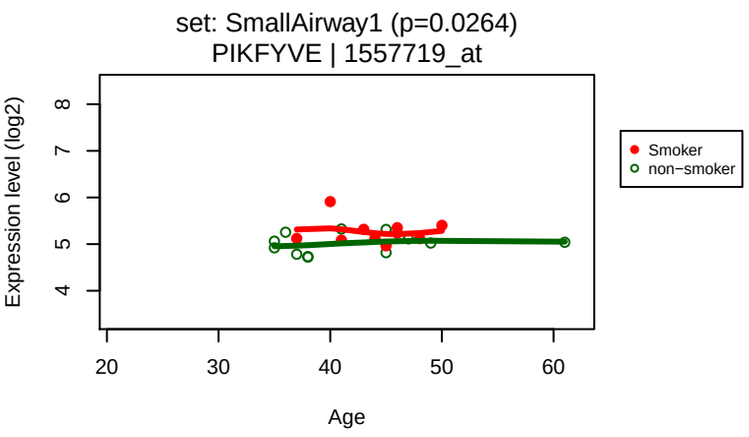
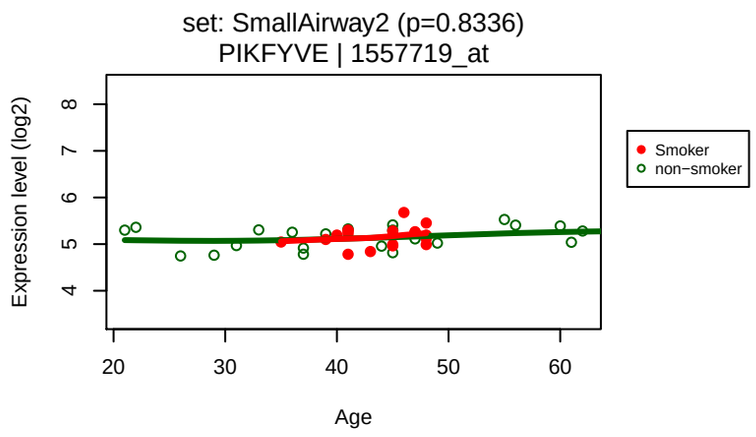
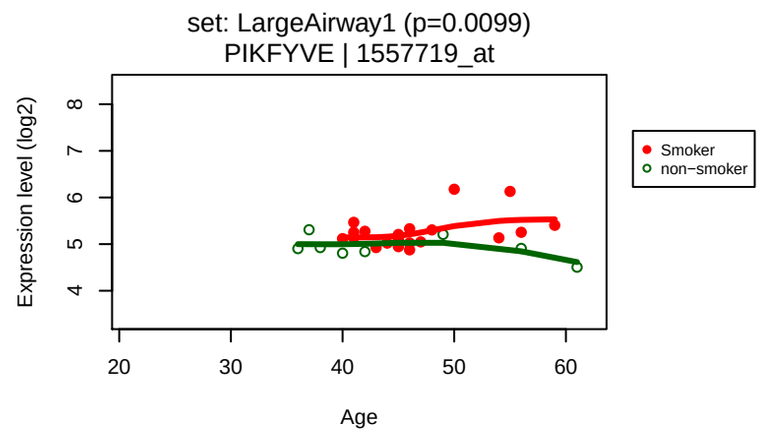

Gene expression of smokers and nonsmokers
gene: PIKFYVE | 1557719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0264
p=0.8336
p=0.6656
p=0.0099
p=0.0275
p=0.4339
p=0.0095
p=0.1531
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

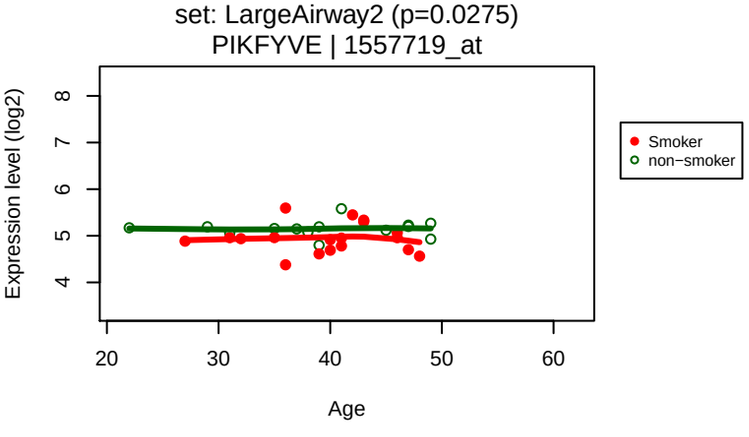

Gene expression ratio (smokers/nonsmokers)
gene: PIKFYVE | 1557719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0264
p=0.8336
p=0.6656
p=0.0099
p=0.0275
p=0.4339
p=0.0095
p=0.1531
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

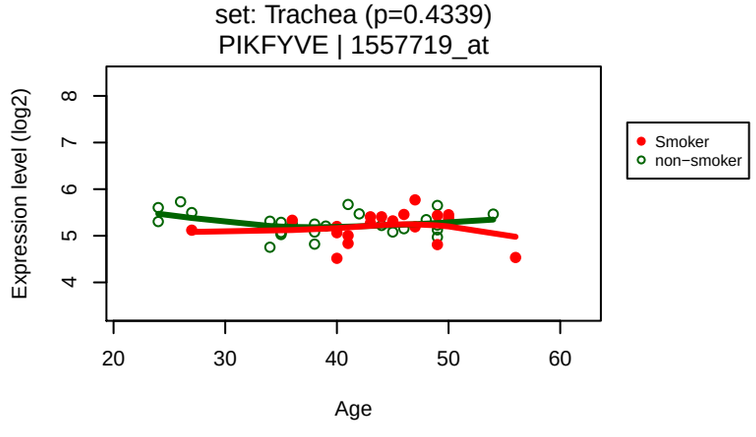

Gene expression ratio in Males
gene: PIKFYVE | 1557719_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9856
p=0.0557
p=0.7238
p=0.0916
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIKFYVE | 1557719_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6336
p=0.3195
p=0.6245
p=0.8946
SmallAirway2
LargeAirway2
Trachea
Macrophage2

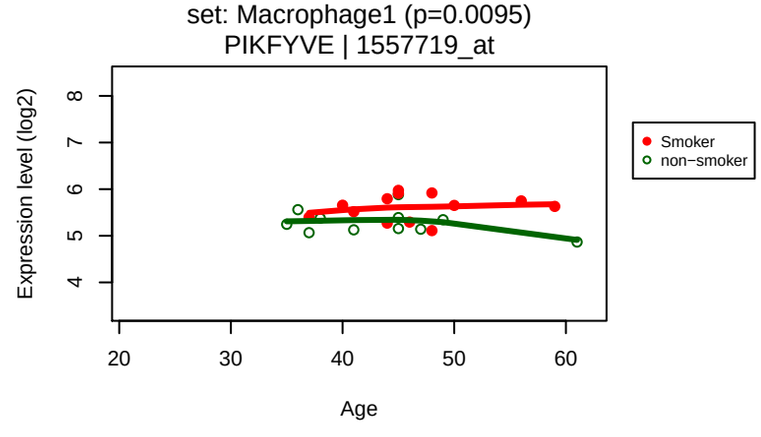

set: SmallAirway1 (p=0.0264)
PIKFYVE | 1557719_at
Expression level (log2)
Age
Smoker
non−smoker

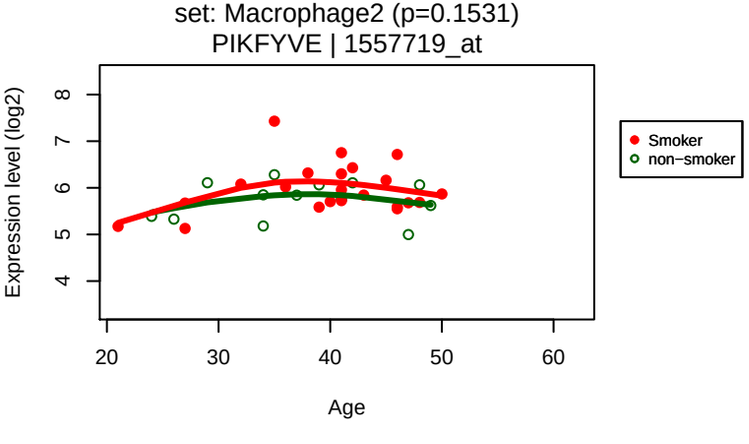

set: SmallAirway2 (p=0.8336)
PIKFYVE | 1557719_at
Expression level (log2)
Age
Smoker
non−smoker

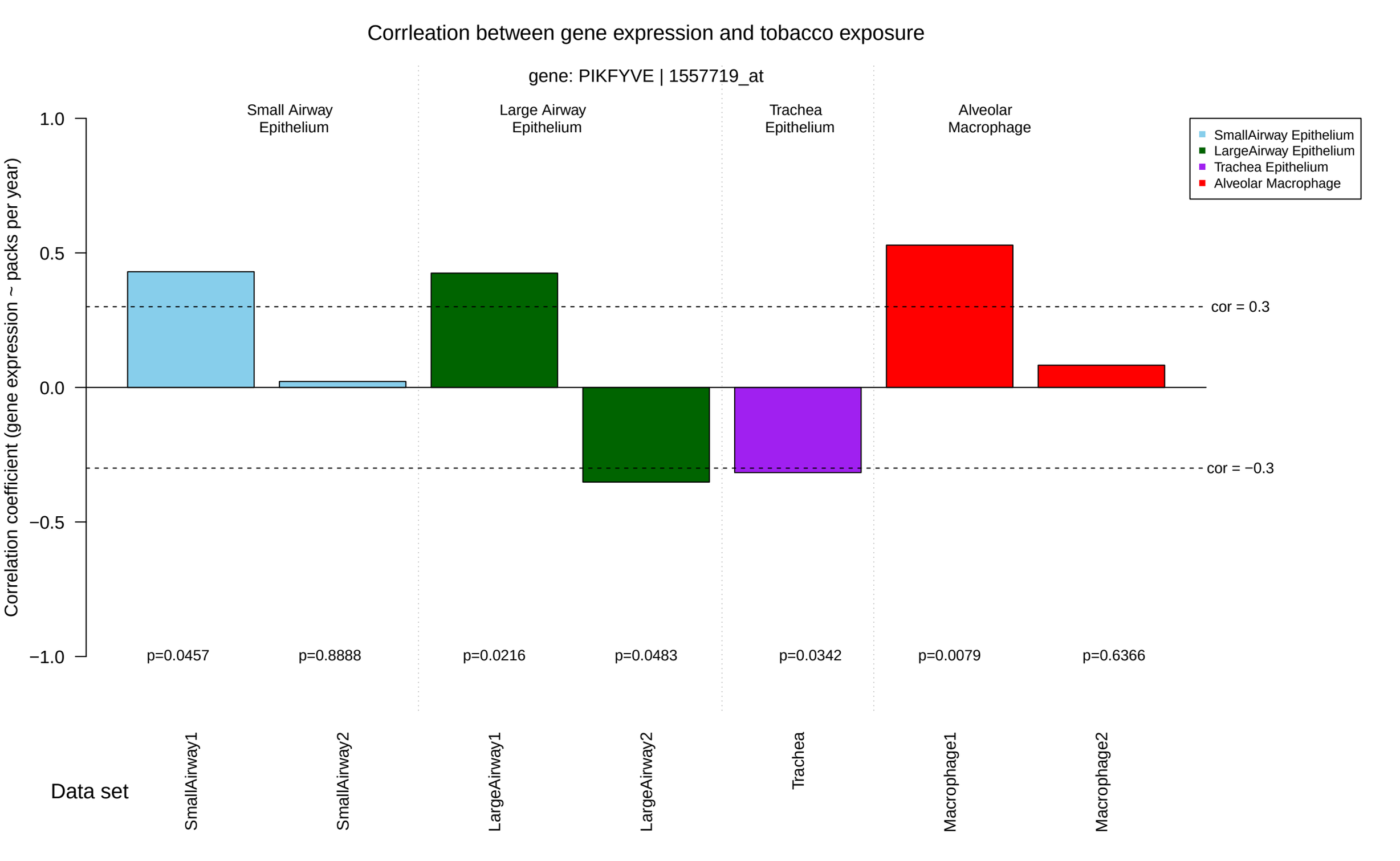

set: LargeAirway1 (p=0.0099)
PIKFYVE | 1557719_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0275)
PIKFYVE | 1557719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4339)
PIKFYVE | 1557719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0095)
PIKFYVE | 1557719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1531)
PIKFYVE | 1557719_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIKFYVE | 1557719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0457
p=0.8888
p=0.0216
p=0.0483
p=0.0342
p=0.0079
p=0.6366
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2